User manual — Field Box BT

App v0.11.0 · box firmware 0.4.1-bt · for teachers, no technical background needed

1. About this manual

Field Box is a field kit for environmental measurements, used by teachers and pupils on school trips and route walks. The kit has two parts: **the box** (a small sensor unit carried in a backpack) and **the phone** with the **Field Box BT** app (kept in a pocket). This manual describes both, function by function: what every button does, what every message means, and what to do when something does not work.

Who it is for. Teachers and pupils using the kit in the field. No technical knowledge is assumed. Installing the app and updating the box are described separately, in the installation guide (**Install_EN.pdf**).

How it is organised. Chapters 2 and 3 introduce the box and the phone. Chapter 4 describes every screen of the app. Chapter 5 is the field day, step by step. Chapters 6 and 7 say which files result and what the reports contain. Chapter 8 is the troubleshooting table: symptom, probable cause, what to do. At the end: specifications, glossary and the one-page field card.

Conventions.

- Button and menu names are in **bold**: **Start**, **Connect to a box**.
- Messages shown by the app are written `like this`: `The clock is not set yet.`
- The app is **English only**, whatever the phone's language.
- `BOX_xxxx` stands for the name of your box (for example `BOX_3C3C`).

Versions covered. App **0.11.0** and box firmware **0.4.1-bt** (0.3.1-bt works too, without the optional sensors). Boxes still on firmware 0.2.0-bt keep working, with the differences noted in chapter 9.

2. The box

2.1 What it is

The box is a small autonomous logger. It has three sensors, an internal memory, a status LED and a USB-C socket for power. It talks to the phone over **Bluetooth**; it has no screen and no buttons. From the moment it receives power it measures and writes **one row every 5 seconds** into its memory, whether or not any phone is connected.

2.2 The sensors and what they measure

Sensor	Measures	Unit	Shown in the app as
BME280	air temperature	°C	**Temperature**
BME280	relative humidity	%	**Humidity**
BME280	atmospheric pressure	hPa	**Pressure** (plus an estimated altitude, in metres)
SCD41	carbon dioxide (CO_2)	ppm	**CO_2**

Sensor	Measures	Unit	Shown in the app as
VEML7700	light	lux	**Light**

The CO_2 sensor produces a new value every 5 seconds; the others, every second. A sensor that does not answer leaves its cell empty in the files and shows – in the app — the box carries on with the others.

Optional sensors (firmware 0.4.1-bt). A box can also carry extra sensors; the box finds them by itself at power-up, and a missing one changes nothing. Their values appear only in the app, on the **Live** tab (4.2); the box's files keep the same columns.

Sensor	Measures	Unit	Shown in the app as
BMV080	particles PM1, PM2.5, PM10	µg/m³	**Particles PM1 / PM2.5 / PM10**
SGP41	volatile organic compounds, nitrogen oxides	index 1–500	**VOC index**, **NOx index**
SGP30	total VOC; CO_2 estimated from it	ppb; ppm	**TVOC**, **eCO_2 (estimate)**
STEMMA soil probe	soil moisture; soil temperature	raw value; °C	**Soil moisture**, **Soil temperature**
BH1750	light (only when there is no VEML7700)	lux	**Light**

The VOC and NOx indices compare the air with that of the last hours (100 = usual), they are not concentrations. The eCO_2 of the SGP30 is an estimate, not a CO_2 measurement like the SCD41's.

The CO_2 bands used in the charts and the report are those of the Magic Boxes lesson:

Band	CO_2 (ppm)	Colour
Excellent	below 800	green
Good	800–1200	yellow-green
Moderate	1200–2000	amber
Poor	2000–5000	red
Dangerous	5000 and above	purple

The temperature bands (for the outdoor lesson, not a comfort standard):

Band	Temperature	Colour on the chart
Cold	below 5 °C	red
Cool	5–15 °C	amber
Comfortable	15–28 °C	green
Hot	28–35 °C	amber
Very hot	35 °C and above	red

2.3 Power

The box is powered through its **USB-C socket**, from a powerbank or a phone charger. Push the plug in firmly; a small red light inside shows it has power. It has no battery of its own and no power button: while it is powered it runs; when you unplug it, it stops.

How long a powerbank lasts depends on its capacity; on Bluetooth the box draws noticeably less than the old Wi-Fi version did. For one field day an ordinary phone powerbank is enough — charge it the evening before.

2.4 The status LED

LED	Meaning
solid for 2 seconds right after power-up	the sensors started correctly
slow blink (once every half second)	a recording is running (**Start** was pressed in the app)
very fast blink (10 times a second)	the box's storage is full — see 2.5 and chapter 8
off	the box is running normally with no recording started (the continuous log is written anyway)

If the LED does **not** light at all after power-up, the box did not start or a sensor is not answering: unplug and plug the cable again.

2.5 The box's storage

The box has about **2.5 MB** of file storage. A log row is about 60 bytes, so at one row every 5 seconds the storage holds **about 150 hours** (some six days) of continuous logging.

The box writes two kinds of file:

- the **continuous log** (`log_<day>_<time>.csv`): a new file at every power-up and at midnight, one row every 5 seconds;
- the **points file** (`points_<day>.csv`): one file per day, one row for every **Read** pressed.

When free space drops below **100 kB**, the box **deletes its oldest continuous log by itself** to make room. Points files are never deleted automatically. If there is no old log left to delete, the box closes the current log, the LED blinks fast and the app shows … `kB free on the box — full; the box has stopped logging`; reads (**Read**) are still accepted as long as one more row fits.

The app warns early: below **300 kB** free, the **Point** tab shows `n kB free on the box — running low; download and delete old files on Files`. Download (**PDF report** or **Send all** fetch the day's log by themselves) and delete old files on **Files**.

2.6 The Bluetooth name

Every box announces itself over Bluetooth as **`BOX_xxxx`**, where `xxxx` are four characters unique to that box (the same name is written on the box). This name goes into every file downloaded from the box and into the photo file names, so files from different boxes never mix.

2.7 The box's clock

The box has **no battery-backed clock of its own**. At every power-up it does not know the time; it learns it from the phone, automatically, the moment they connect. Until then:

- the rows measured are kept in temporary memory (at most one hour) and written with the correct time as soon as the phone connects;
- **Read** and **Start** are refused (`The clock is not set yet.`);
- on the **Boxes** tab the clock shows **`(NOT SYNCED)`** in red.

The time stays set as long as the box is powered, even if the Bluetooth link drops. It is lost only when the cable is unplugged.

2.8 The box without a phone

The box logs **whenever it is powered**, with or without a phone. You can leave it on all day in a classroom; the next time a phone connects, **Files → PDF report** fetches the day's log by itself and makes the day's report (the **Box day** pages, chapter 7). The only condition is that at some point that day a phone connected to set its clock; the rows from before (at most one hour) are recovered, older ones are lost.

3. The phone

3.1 Requirements

- A phone or tablet with **Android 10 or newer**.
- **Bluetooth** (for the box), **GPS** (for the route and the positions of reads) and a **camera** (for photographs). Without GPS or without a camera the app still works, minus those functions.
- No internet is needed in the field. Internet (Wi-Fi at home) is only used to prepare the map (chapter 5).

3.2 Installation

The app is installed from a file, `fieldbox-bt-0.11.0.apk`, not from the Play Store. The steps, with screenshots and the Android warnings that appear, are in the installation guide (**Install_EN.pdf**). The app appears on the phone as **Field Box BT**.



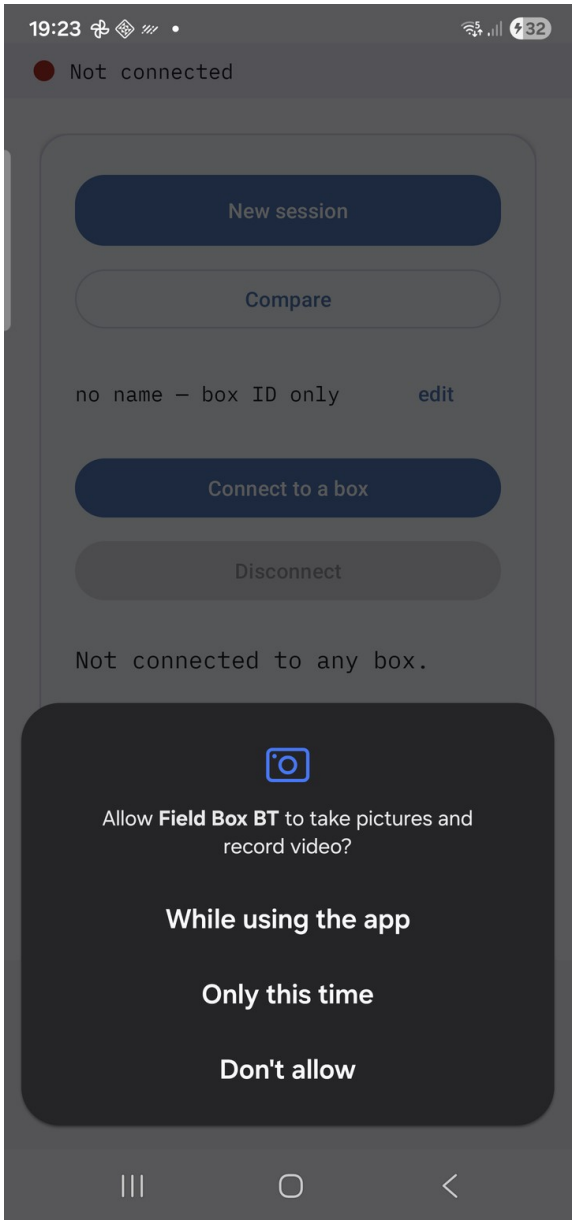
The Field Box BT icon, as it appears on the phone

3.3 Permissions asked and why

At first launch the app asks, one after another, for the permissions below. Grant them all; without one, that function is missing.

Permission	Used for	If you refuse
Location (choose the *precise* option)	the GPS route, the position of every read and every photo	reads are saved without coordinates; the route is not recorded; Live shows `Route: not running — check the Location permission…`
Nearby devices / Bluetooth	talking to the box	the app cannot connect to any box: `Bluetooth permission is required to talk to the box. Tap Connect again and allow it.`
Camera	the photographs at stations	`Camera permission is needed to photograph a station.`
Notifications (Android 13 and newer)	the permanent notification with elapsed time and the **Read now** button	you do not see the notification; the recording runs anyway

Permissions can also be granted later from Android settings: **Settings → Apps → Field Box BT → Permissions**. If Location was granted only after **Start** was pressed, stop and start the recording again (or tap the `Route: …` line on Live) so the route starts.



The Android permissions screen

3.4 Pairing with the box (once)

The phone and the box must be **paired** once, from the phone's Bluetooth settings: **Settings → Bluetooth → Pair new device → `BOX_xxxx`**. Do it at home, not in the field. A box that has never been paired **does not appear** in the app's list — this is the most common cause of "the app can't see my box". The steps with screenshots are in the installation guide.

Pair the phone with **every** box it might use.

3.5 The tabs at a glance

The app has five tabs, in the bottom bar:

Tab	What it is for
Boxes	connecting to the box; its state (clock, storage); the profile
Live	the sensors' current values, with trend arrows, and the charts of the last 50 minutes
Point	the measurement day: Start, the station list, Read, photographs, Stop, report
Map	today's route, coloured by one sensor; stations and photographs on the map
Files	the files on the box; the reports; Compare; the plan template

Secondary screens, opened from the tabs: **Profile**, **Station plan**, **Compare**, the camera.

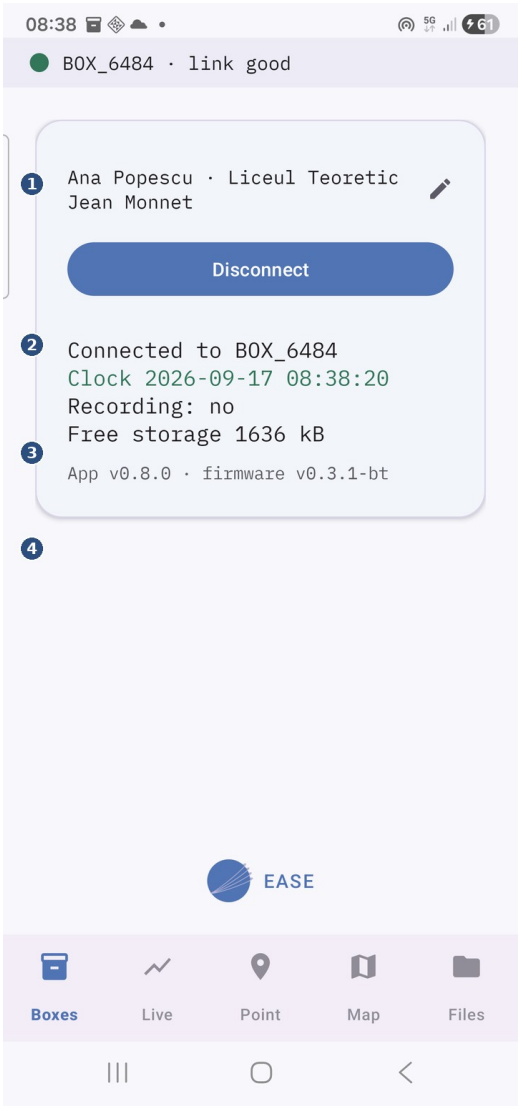
The link strip. At the top of every tab a strip shows the state of the link to the box, so you never have to go back to Boxes:

What you see	What it means
🟢 **`BOX_xxxx · link good`**	connected, the box answers immediately
🟢 **`BOX_xxxx · link ok`**	connected, answers are slower — usually fine
🟠 **`BOX_xxxx · link poor`**	an answer was late or missing in the last 10 seconds: you are at the edge of range — bring the box closer to the phone
🟣 **`BOX_xxxx · link none`**	just connected; the link has not been judged yet (a few seconds)
🟠 **`Connecting…`**	you tapped a box and the app is connecting
🟠 **`Reconnecting to BOX_xxxx…`**	the link dropped; the app is restoring it by itself
🔴 **`Not connected`**	no box — go to **Boxes** or **Point** and choose the box

The link is judged by how quickly the box answers the app's questions (once a second): under 0.3 s = `good`; over 1 s, or a missing answer = `poor`. Bluetooth offers no "signal bars", so this is the only signal indicator.

4. The screens, function by function

4.1 Boxes



The app's start screen.

① **The profile row** — who carries the phone; tap it (the pencil) to change it.

② **Disconnect** — while connected; otherwise this is **Connect to a box**.

③ **The box's status** — which box you are connected to, its clock (green = set), whether it is recording, free storage.

④ **Versions** — app and box firmware; a long press opens the link log (testing only).

The profile row. At the top of the card is who carries the phone, as entered in the profile (for example `Ana Popescu · Liceul Teoretic Jean Monnet`) or `no name – box ID only` if nothing was entered. A small pencil at the end of the row is the hint that **tapping the row** opens the **Profile** screen (4.7).

Connect to a box. Opens the `Choose a box` dialog with the list of boxes **paired** with this phone. Tap yours. It is not a search: an unpaired box does not appear. If no box is paired at all, the message `No paired box found. Open Bluetooth settings, pair with the BOX_XXXX device (it appears when the box is powered), then come back and tap Connect.` appears with a **Bluetooth settings** button that opens the phone's Bluetooth settings. While you are connected the **Connect to a box** button is hidden.

Disconnect. Closes the link and **forgets** the box: the next connection shows the list again. Tapped by mistake, it loses nothing — reconnect.

The status lines (under the buttons), depending on the situation:

- `Not connected to any box.` — no box.
- `Connecting…` — in progress.
- `Reconnecting to BOX_xxxx…` — the link dropped; **Connect to a box** stays enabled so you can pick another box if needed.
- When connected, four lines: `Connected to BOX_xxxx`; `Clock 2026-09-16 10:29:43` — in green if the clock is set, followed by **`(NOT SYNCED)`** in red if not (see 2.7; fixed by disconnecting and reconnecting); `Recording: yes` or `Recording: no` — whether **Start** was pressed; `Free storage n kB` — free space on the box.
- If connecting failed, a sentence you can act on and, beneath it in small type, the technical text of the error: `The box is not answering. Check it is switched on and within a few metres, then tap`

`Connect again.`; `Bluetooth is switched off on this device. Turn it on, then tap Connect again.`; `Could not connect. Try again; if it keeps failing, switch the box off and on.`

The versions line. At the bottom: `App v0.11.0 · firmware v0.4.1-bt` (with – instead of the firmware until a box connects). These are the two numbers any support conversation needs.

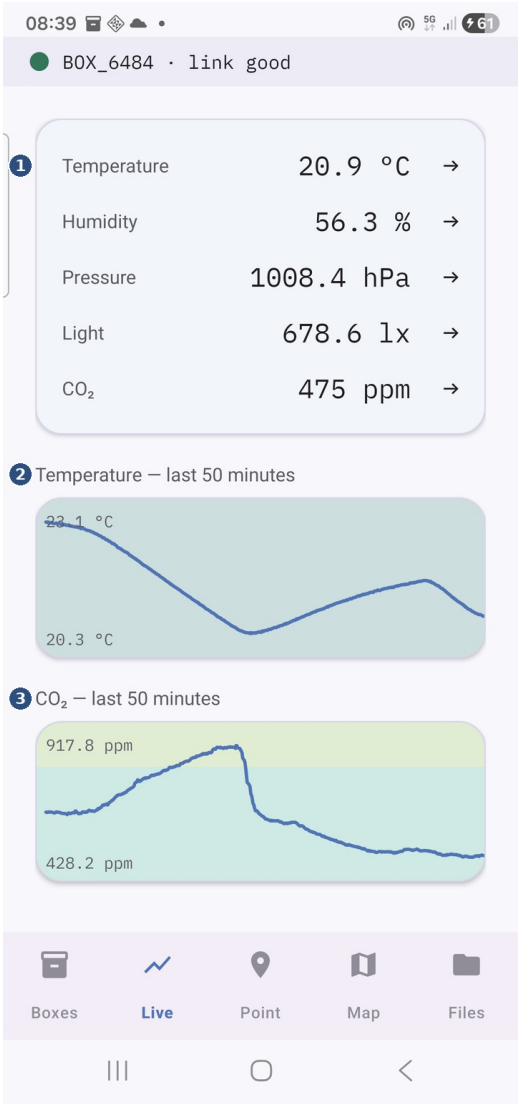
A new version. When the phone has internet, the app asks GitHub at most once a day whether a new version is out (one request, nothing about the school or the measurements). If one is, a card under the box card says for example 🎁 `FieldBox 0.9.1 is out` with a **Download** button that downloads the new app in the browser; install it over the old one, as in the installation guide. Without internet nothing appears.

The link-drop log (for testing, not for pupils): a **long press** on the versions line opens the `Link drops` dialog. It shows `Drops: none recorded` or `Drops: 2 · longest 14 s` (possibly `· 1 needed a tap` = one needed you to press Connect yourself), then the last 12 events, one per line: the time, then `back in 6 s` (healed by itself), `back after 14 s (tap)` (healed manually) or `down…` (still down). **Clear drop log** empties the list. A few short drops a day are normal.

The link notification. While a box is connected, the notification bar holds `Field Box — connected to BOX_xxxx` / `Keeping the Bluetooth link to the box alive in the background.` Leave it there: it keeps the link alive with the screen off. The phone's mobile internet keeps working.

Automatic reconnection. If the link drops (the box goes out of range, or blinks off for a moment), the app tries to restore it by itself, once a second, whichever tab you are on. Nothing to do; the box keeps logging meanwhile.

4.2 Live



The sensors' values right now and two charts. Works without **Start**: useful for a demonstration in class.

① **Current values** — temperature, humidity, pressure, light, CO_2, each with its arrow: ↑ rising, ↓ falling, → steady, … not known yet (first 40 s).

② **Temperature, last 50 minutes** — the coloured bands behind it: cold, cool, comfortable (green), hot, very hot.

③ **CO_2, last 50 minutes** — the lesson's bands: under 800 excellent, up to 1200 good, up to 2000 moderate, up to 5000 poor, above — dangerous.

Without a box. If no box is connected the screen shows only `Connect to a box first, on the Boxes tab.` If the link drops while you are watching: `Connection lost — reconnecting… the box may be out of range.` The values come back by themselves when the link is restored.

The values. Five cards — **Temperature** (°C), **Humidity** (%), **Pressure** (hPa), **Light** (lx), **CO_2** (ppm) — refreshed once a second. – means that sensor has not produced a value yet (CO_2 needs a few seconds after power-up; see chapter 8).

The trend arrows. After every value there is an arrow:

Arrow	Meaning
↑	rose compared with 30 seconds ago
↓	fell
→	stayed put (change below the noise threshold)
… (faint)	not known yet — during the first **40 seconds** after connecting, or after a gap

The comparison is between the mean of the last 10 seconds and the mean of 10 seconds from 30–40 seconds ago. Changes smaller than the sensor's noise count as "stayed put": **30 ppm** for CO_2, **0.2 °C** for temperature, **1.5 %** for humidity, **0.5 hPa** for pressure, **10 lx or 20 %** of the value for light (whichever is larger).

Extra sensors. On a box with optional sensors (2.2), a card under the five values lists only the ones fitted, refreshed once a second. Right after power-up the SGP41 and SGP30 show `warming up` (about 45 s for the SGP41, 15 s for the SGP30; the VOC index settles over the first minutes). If the particle sensor is covered by the

lid, a hand or an object, the card says so and shows – for PM until its view is clear. Soil moisture is the probe's raw number: about 200 in dry air, up to about 2000 in water, and different for every probe, so compare readings of the same probe. On a box without extra sensors the card does not appear.

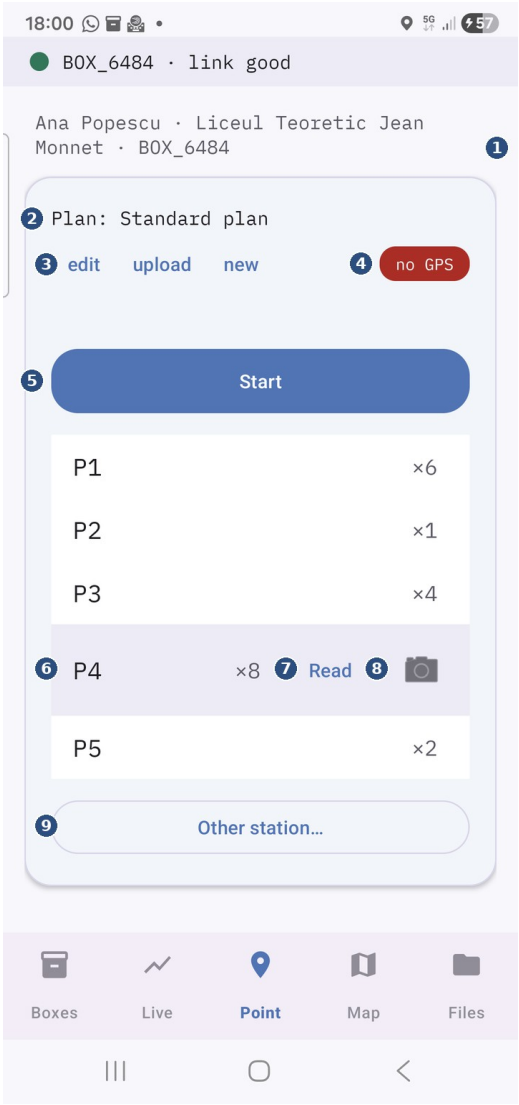
The charts. `Temperature — last 50 minutes` and `CO₂ — last 50 minutes` show the history the box keeps (one point every 5 seconds, the last 50 minutes) — so they appear filled right after connecting, not from zero. Behind the line the lesson's bands are painted (2.2): for CO_2 green / yellow-green / amber / red / purple, for temperature red / amber / green / amber / red. Vertical ticks mark the **Read** moments. The text `waiting for data` in place of the chart means the history has not arrived yet (the first seconds) or the box has firmware that is too old (chapter 8).

The Route line. Shown only while a recording is running; it says what the phone's GPS route is doing:

Text	Meaning / what to do
`Route: waiting for the first GPS fix…`	normal at the start; get out under an open sky
`Route: no point yet · n GPS fixes rejected (last ±x m, limit 100 m)`	the phone delivers positions, but all worse than 100 m — indoors or next to buildings; wait outside
`Route: n points recorded`	working well
`Route: n points · m skipped (weak GPS)`	m positions were rejected (inaccurate, over ±100 m, or too close, under 3 m); normal indoors or under tree cover
`Route: not running on the phone (the box is recording) — tap here to resume the route`	the phone's route stopped (usually after an app update or when Android closed the app); **tap the message** — the route resumes, in a new file; the box lost nothing
`Route: not running — check the Location permission, then stop and start the recording`	the Location permission is missing; tap the message (it asks for it), then stop and start from Point
`Route: not running — the system refused the start: …`	Android refused the location service; the text after the colon is the technical reason; stop and start the recording
`Route: off — this device has no GPS receiver`	tablet without GPS: there will be no route; reads work without coordinates

The recording is started and stopped from **Point** (Start / Stop); the elapsed time is in the notification (4.9).

4.3 Point



The measurement day's screen. It shows only what still needs an answer, then the station list. Chapter 5 walks through it in the order of a day; here are all the controls.

① **The header** — who carries the phone and which box it talks to.

② **The active plan** — the list of stations you walk.

③ **edit · upload · new** — change the plan, load one from a CSV, or start the next pupil's walk.

④ **The GPS pill** — red `no GPS`, amber `weak`, green `GPS n m`.

⑤ **Start / Stop** — starts and stops the day's recording.

⑥ **The current station** — the highlighted row; tap another row to make it current. The number on the right: how many times it was read.

⑦ **Read** — one reading at the current station (all sensors + the GPS position).

⑧ **The camera** — a photograph tied to the current station.

⑨ **Other station…** — add a station that is not on the plan; a long press on it removes it.

The header. The first line, of the form `Ana Popescu · Liceul Teoretic Jean Monnet · BOX_3C3C`, says who carries the phone and which box it talks to. `no profile` means the profile is empty; `no box` that no box is connected.

The Box card *(only while no box is connected)*. The title `Box`, the list of paired boxes — tap yours (it connects directly, without going through Boxes) — and, below, `Choose the paired box` or `Connecting…`. If the Bluetooth permission is missing, the card asks for it and says `Bluetooth permission is required to talk to the box. Tap Connect again and allow it.`

Continue without a box. Hides the Box card for a **photographs-only** walk: the station list appears, the camera works, **Read** and **Start** stay disabled until a box connects (the status line says `Connect a box to read — Photograph works without one.`). If a box connects later, the choice cancels itself; if the link is lost for good, the Box card comes back.

The Profile card *(only the first time)*. The title `Profile`, the explanation `The name appears in the report and in file names. Without a name, the session is identified only by the box's ID.` and two buttons: **Set profile** (opens the Profile screen, 4.7) and **Skip** (no name; the card does not come back). The profile can be filled in any time later, from Boxes or from **new**.

The plan line. `Plan: Standard plan` — the name of the active station plan — followed by three small buttons:

- **edit** — opens the **Station plan** screen (4.6);
- **upload** — loads a plan from a CSV file (see 4.6, "The plan from a file");
- **new** — the next pupil: the **New walk** dialog (below).

The GPS pill. Under the plan line, a coloured pill with the positioning state:

Pill	Meaning
🟢 `GPS 8 m`	a good fix, accuracy of at most **15 m**
🟠 `GPS 32 m (weak)`	a weak fix (over 15 m) — it is used; keep going; it improves under an open sky
🔴 `no GPS`	no fix yet — at the start, indoors, with Location off or without the permission

GPS **never blocks** a read or a photo: without a fix, the read is saved with the station and the time, but without coordinates. Once, when the tab opens, one of these may appear: `Location is switched off on this device. Readings will be recorded without coordinates.` (turn Location on in the quick settings), `Location permission refused. Readings will be recorded without coordinates.` (grant it in Settings) or `This device has no GNSS receiver. Readings will be recorded without coordinates.` (a tablet without GPS).

The status line (under the pill). Empty when all is well. Otherwise:

- `n kB free on the box — running low; download and delete old files on Files` — under 300 kB free on the box;
- `n kB free on the box — full; the box has stopped logging` — storage full (2.5);
- `Reconnecting to BOX_xxxx…` / `Link to the box dropped — reconnecting…` — the link dropped; the app is restoring it;
- `Connect a box to read — Photograph works without one.` — you are working without a box;
- after a **Start** the box refused, the reason: `The clock is not set yet.` (2.7), `The storage is full.` (2.5), `The box refused the request.` or `Could not reach the box.` The message stays until the next successful Start.

Start. Starts the recording: marks the start in the box's log, starts the phone's own series (one row every 5 seconds with the box's values), the GPS route and the permanent notification (4.9). The button is enabled only with a box connected. The plan's first station becomes the current station, if none was yet.

Stop. Replaces Start while the recording runs. Stops the series, the route and the notification, marks the end in the box's log, writes the day's bundle (the `session_….json` and `.gpx` files, chapter 6) and shows the **Done** card. It works even if the box has meanwhile disconnected or if the phone has no GPS.

The station list. One row per station in the plan, plus those added today. On each row: the station's name and, once it has been read, how many times (×1, ×2 …). **The current row** is highlighted (lavender background) and only it has the **Read** and camera buttons. The current row is the one you tapped last; it **never advances by itself** — a group reads the same station several times before moving on.

- **Tap a row** to make it current ("this is where we are now").
- **Read** — one read at the current station: the box writes a row in its points file (with the phone's GPS position, if any), the phone writes its own copy, the counter grows and the phone **vibrates** briefly. A Read

pressed before Start starts the recording by itself. If the box refuses: `The box could not record this point — check the clock is synced and there is free storage.`; if it does not answer: `Could not reach the box.` The button is disabled without a connected box.

- **The camera** (the icon on the current row) — opens the capture screen (4.10). Disabled while a photo is in progress.
- **Other station…** (at the end of the list) — a dialog with a `Station name` field: adds a station that is not in the plan, puts it in the list and makes it current. Usable any time, including mid-walk. It does not change the plan.
- **Long press** on a station added with Other station… — the dialog `Remove P6 from today's list? Its readings stay in the files.` with the **Remove** button takes it off the list; reads already made stay in the files. The plan's own stations cannot be removed here (edit them in Station plan).

The encouragement messages. At 1, 5, 10, 20 and 40 reads in one day the app shows a `Field Box` dialog with a short message from the team that built the kit. Tap OK; it has no effect on the data.

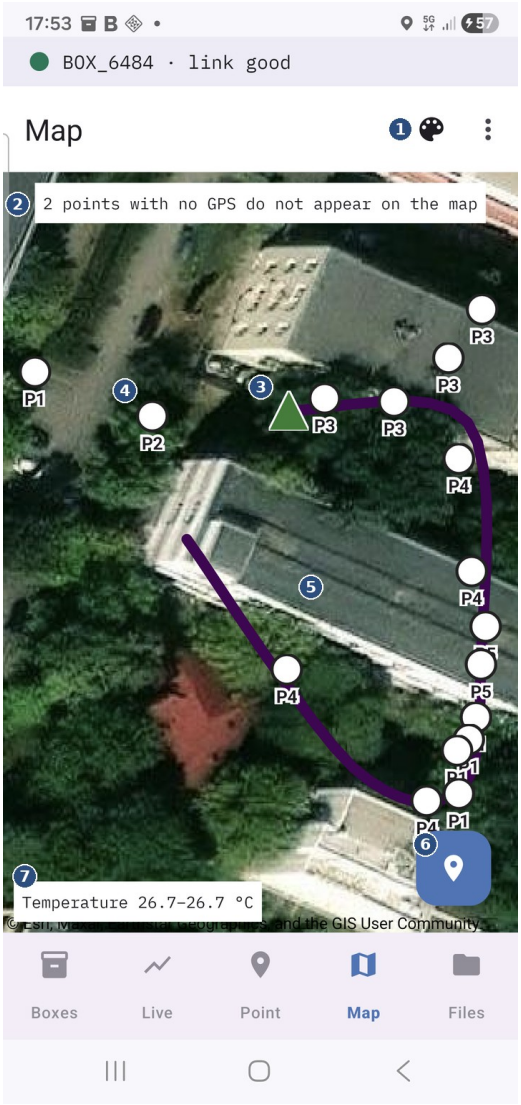
new — the next pupil. The `New walk` dialog has the fields `Name`, `Group / school (optional)` (prefilled with the current profile) and the `Station plan` list. **Start over** saves the chosen profile and plan, clears the counts, the current station and the stations added today, and hides the Done card. If there are reads today already, it asks first: `Counts for n stations will be cleared — the files on the phone and the box stay.` — the files are untouched; only the on-screen counts start from zero.

The Done card *(after Stop)*. A one-line summary — `12 points · 1.84 km · 3 photos` — and five actions:

- **Map** — opens the Map tab on today's route.
- **PDF report** — makes the day's report (chapter 7) and opens the share sheet. With a box connected it first fetches its log (the button reads `Downloading the box log… n kB`). The report made from here **does not contain the map**; for the report with the map, open the Map tab first, then make the report from **Files**.
- **Send all** — makes a ZIP with everything (chapter 7) and shows `Package saved` with a **Send** button that opens the share sheet (WhatsApp, e-mail, Drive…). On failure: `The package could not be created`.
- **Continue measuring** — brings the list back with its counts; the next **Start** records a second route in the same day, and the next Stop rewrites the day's bundle with both.
- **New walk (next pupil)** — the same dialog as **new**.

If there is nothing to report for the day, the summary reads `Nothing to report yet.` and PDF report / Send all are disabled.

4.4 Map



outline marks the **start** (`start 10:29`).

Today's route on a map, coloured by one sensor, with the stations and the photographs. It reads only what is already on the phone — it never asks the box — and refreshes by itself every 30 seconds, so it can be watched while walking.

① **The palette** — choose the sensor that colours the route (temperature, humidity, pressure, CO_2, light or none); next to it the ⋮ menu with the satellite view and **Archive today**.

② **The banner** — how many reads have no GPS position and so are not on the map.

③ **The green triangle** — where the recording started.

④ **The named circles** — the stations read; a station read several times is one circle, `P3 ×4`; tap it for the values.

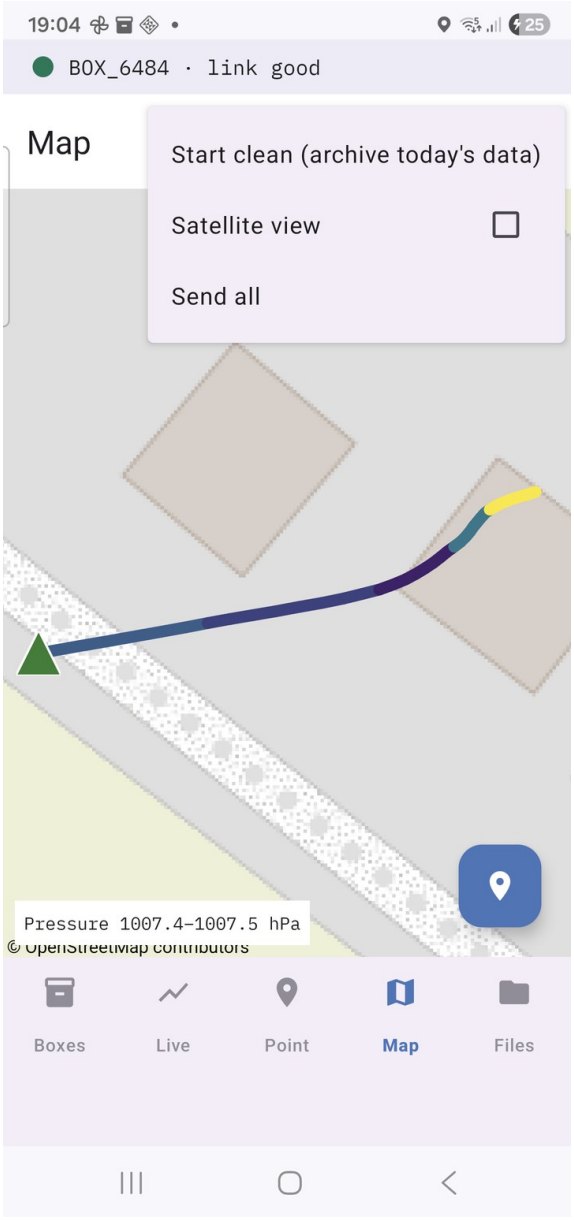
⑤ **The route** — the path walked, coloured by the chosen sensor.

⑥ **The position button** — centres the map on where you are.

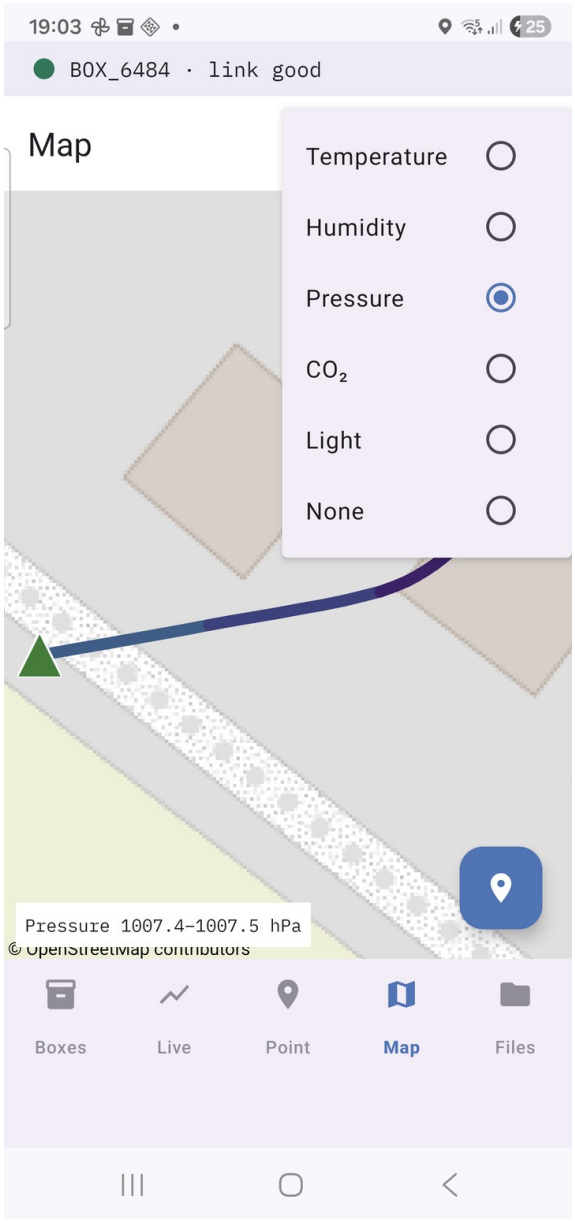
⑦ **The legend** — the chosen sensor and its range along the route (in this version it sits in the top bar, next to the palette).

The route. A continuous line through the recorded positions, **coloured by the chosen sensor** (temperature by default): from dark purple (low values) to yellow (high values). The legend in the top bar gives the sensor and its range, with the unit, for example `Temperature 18.2–24.6 °C` (the range is between the 5th and 95th percentile of the values along the route, so two extreme values do not wash the colours out). Stretches with no sensor value within ±60 s stay navy. A green marker with a white

⋮ → **Colour route by.** Picks the sensor: **Temperature** (default), **Humidity**, **Pressure**, **CO_2**, **Light** or **None** (a plain navy line, no legend). The choice is remembered.



The ⋮ menu on Map



The Colour route by submenu

⋮ → **Satellite view.** Swaps the street map for satellite imagery. The tick is remembered. Map tiles seen once (on Wi-Fi) stay saved on the phone, separately for each view — prepare the area in advance in the view you will use.

The stations. Every station that was read appears as a **white circle** with its label. A station read several times is **one circle**, labelled `P3 ×4`, placed at the mean position. Tap the circle: a bubble titled `P3 · 4 reads · 11:30:26–11:42:10` (or `P3 · 11:30:26` for a single read) with the **averages** of the reads: `Temperature: … °C`, `Humidity: … %`, `CO₂: … ppm`, `Pressure: … hPa`, `Altitude: … m`, `Light: … lux`. Tap the bubble or the map to close it.

The photographs. Every photograph appears as a **thumbnail** where it was taken (it shrinks as you zoom out). Tap it: the bubble shows the photo, `P3 · 11:30 · 12 m from the point` (the station it belongs to — the

nearest read point — and the distance to it) and the values measured at that point. If the photo was deleted or moved out of the gallery, its file name and `the photo was not found` appear instead.

Centre on me. The round button in the corner recentres the map on the phone's current position. Without a fix: `No GPS fix yet — open the Point tab to start locating.`

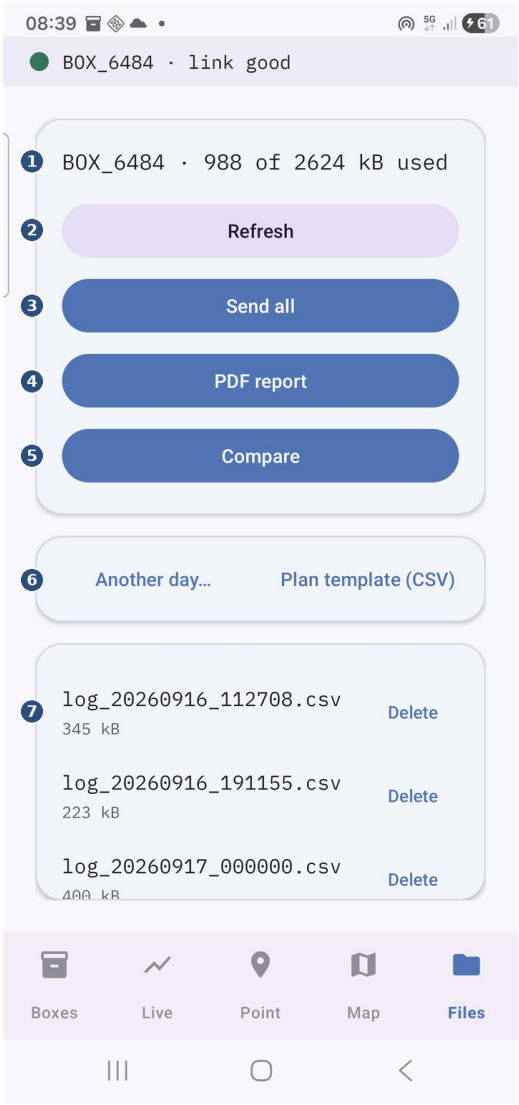
Messages on the map.

- `No measurements today. The route appears after Record, points after Read now.` — nothing yet today; the map stays on Romania.
- `n points with no GPS do not appear on the map` — n reads were made without a fix (indoors); they are in the files and the report, but cannot be drawn.

⋮ **→ Start clean (archive today's data).** For a second session on the same day (another class in the afternoon, or a rehearsal before the real walk). The confirmation says exactly what happens: `Today's files on the phone (route, points, series, photos list, report, box logs) move to Documents/FieldBox/archive/. The map starts empty; nothing is deleted. Continue?` The **Archive** button moves the whole day folder to `Documents/FieldBox/archive/<day>_<time>/`, clears the counts on Point and shows `n files moved to the archive. The map is empty for a new session.` The photographs stay in the gallery (only their list moves, so they no longer appear on the map). Refused while a recording is running: `Stop the recording before archiving today's data.`

The map in the report. The report made from **Files** includes the map as you last saw it on this tab, that day. Open Map once before making the report.

4.5 Files



The files on the box and everything to do with reports.

① **The box's storage** — how much is used of how much there is.

② **Refresh** — re-reads the list of files on the box.

③ **Send all** — builds the day's ZIP (everything the teacher needs) and offers to send it.

④ **PDF report** — today's report; it fetches the box's log by itself.

⑤ **Compare** — the teacher's screen, with several pupils' sessions.

⑥ **Another day...** / **Plan template (CSV)** — a report for another day; a plan template to fill in.

⑦ **The files on the box** — name and size; **Delete** removes them from the box (not from the phone).

The storage line. `BOX_3C3C · 812 of 2560 kB used` — how much of the box's storage is in use — followed by `· RECORDING` while a recording is running. Without a box: `Connect to a box first, on the Boxes tab.` If the box could not be read: `Could not read the box: …`

Refresh. Re-reads the list from the box.

Send all. The same ZIP as on the Done card (chapter 7), from what is already on the phone for today, plus the box's log if it is connected. `Package saved` + **Send**.

PDF report. The report of **today** (chapter 7). With the box connected, it first fetches today's log from it (`Downloading the box log… n kB` on the button; about 20 seconds for a whole day) — keep the phone next to the box. Then `Report saved` + **Send**. If there is nothing today: `Nothing to report for 16 Sep 2026.`; on failure: `The report could not be created.`

Compare. Opens the comparison screen (4.8).

Another day... The report for **another day**. The `Report for which day?` dialog lists the days the phone has anything for, newest first, of the form `16 Sep 2026 — points · series · box log BOX_3C3C` (what exists: points, a series, a box log). Pick the day; its report is made. If there is no day at all: `Nothing on the phone to report yet.` Tracks and photo lists alone do not make a day "with data".

Plan template (CSV). Writes the plan model `plan_model.csv` (the standard plan, one station per line) into Documents/FieldBox: `Saved as Documents/FieldBox/plan_model.csv — one station per line.` + **Send**, to send it to a laptop and fill it in (4.6).

The list of files on the box. One row per file: the name (`log_20260916_102943.csv`, `points_20260916.csv`), the size (`n kB`) and a **Delete** button. The log the box is writing right now has Delete greyed out — it cannot be deleted (or downloaded) while open.

Delete. Confirmation `Delete file` / `Delete BOX_3C3C_log_….csv? This cannot be undone.` → **Delete**. **Files deleted from the box cannot be recovered.** Delete only after the report or Send all has fetched the day's log (downloaded files are in the day folder, chapter 6). Possible messages: `This file is being recorded — stop the recording first.` (press Stop on Point), `Could not delete the file.` (the box did not answer; try again). After deleting a log, if the box immediately opens another one: `Deleted — the box opened a fresh log (it logs whenever it is powered).` — normal, the box logs while powered.

Downloading a single file. There is no separate download button: **PDF report** and **Send all** fetch that day's logs by themselves (and only those that are missing or have changed). A file from another day is downloaded by making that day's report with the box connected (**Another day…**).

4.6 Station plan

A **plan** is an ordered list of stations, with a name. The **`Standard plan`** has the stations **P1, P2, P3, P4, P5** and always exists. The screen opens from **edit** on the plan line (Point tab).

The controls:

- the drop-down at the top — picks the plan to edit (every plan saved on the phone);
- the station list — drag the handle on the right to reorder; swipe a row to the left to delete it;
- the `Station name` field + **+ Station** — adds a station at the end;
- **Save as…** — a dialog with `Plan name`: saves the current list as a new plan (or over one with the same name);
- **Use this plan** — saves the changes under the current name, makes it the active plan and closes the screen.

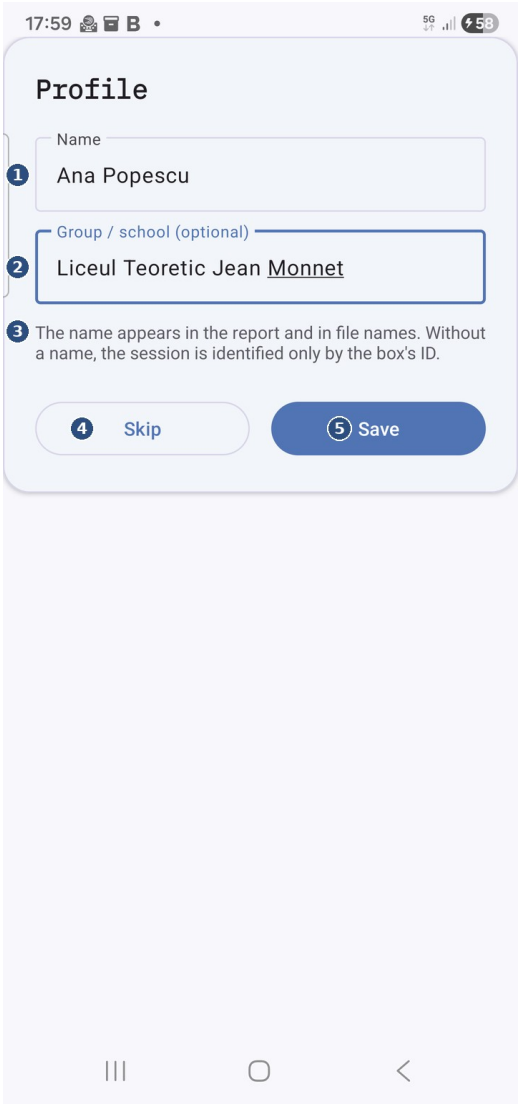
Changes not followed by **Use this plan** or **Save as…** are lost on leaving the screen (the back button). Names are cleaned automatically: leading/trailing spaces, empty rows and duplicates are removed. A plan left with no station at all is replaced by the standard plan.

The plan from a file (upload). The **upload** button on the plan line takes a CSV file (from My Files, WhatsApp, Drive…) and turns it into a plan:

- **one station per line** — or one single Excel row, with commas or semicolons; quotes are ignored;
- a first row that is a header word (`station`, `stație`, `nume`, `name`…) is skipped; any other first row is a station;
- the plan takes **the file's name** (`Parcul_Herastrau.csv` → the plan `Parcul_Herastrau`), is saved on the phone, becomes the active plan and also appears in the Station plan list; at most 200 stations.

Messages: `Plan “Parcul_Herastrau” loaded: 7 stations` (success), `No stations found in x.csv.` (empty file or header only), `Could not read the file.` (the file could not be opened). A model comes from **Files → Plan template (CSV)**.

4.7 Profile



Who carries the phone. It opens at the app's first launch, from the top row of the Boxes tab, from **Set profile** on Point or, in short form, from **new**.

① **Name** — the pupil (or group) carrying the phone.

② **Group / school** — class or school; optional.

③ **The explanation** — the name goes into the report and the file names; without one, the day is identified by the box's ID only.

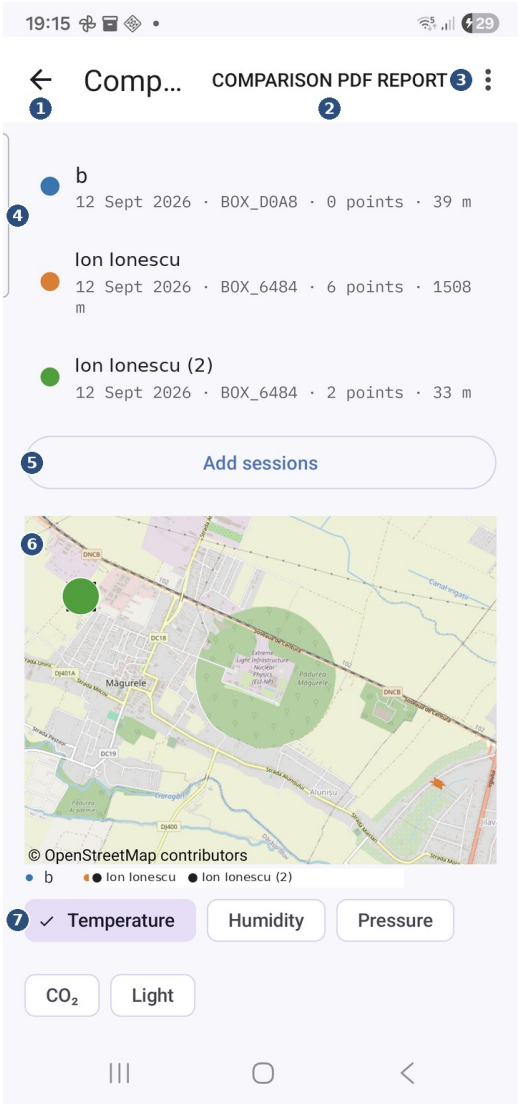
④ **Skip** — go on without a profile (not asked again at start).

⑤ **Save** — saves it; change it any time from Boxes (the pencil) or from **new** on Point.

- `Name` — the pupil's (or group's) name;
- `Group / school (optional)` — the class or school;
- **Save** — saves; **Skip** — leaves the profile blank and stops asking at launch.

Where the name appears: in the Point tab's header, in the top row of Boxes, on the PDF report's cover, in the Compare screen's legend and in file names (`report_<day>_<Name>_<Class>.pdf`, `session_<day>_<BOX>_<Name>.zip`). Without a name, files and the report carry only the box's code.

4.8 Compare (for the teacher)



From **Files → Compare**. Puts up to **6 sessions** — of different pupils — together on one map and one sensor at a time.

① **Back** — to the Files tab.

② **Comparison PDF report** — the comparison report with every loaded session.

③ **The ⋮ menu** — **Clear all** empties the list.

④ **The loaded sessions** — one per row with its colour, day, box, number of points and distance; swipe a row sideways to remove it.

⑤ **Add sessions** — pick the ZIPs the pupils sent (at most 6 sessions).

⑥ **The map** — everyone's route, each in its colour.

⑦ **The compared sensor** — tap a chip; below it, the table and chart by station or by time.

How sessions get in.

- **Add sessions** — pick one or more ZIPs received from the pupils (the ones made with **Send all**). A lone `session_….json` file is accepted too.
- Straight from WhatsApp (or any app): tap the ZIP and choose **"Open with Field Box"** — the session lands here directly.
- The empty screen says: `No sessions loaded. Tap "Add sessions" and choose the ZIPs received from the pupils ("Send all"), or open a ZIP from WhatsApp with Field Box.`
- A 7th session is refused: `Maximum 6 sessions.` — remove one first.
- A file that is not a session: `x is not a Field Box session.`; a broken one: `Could not import x.`

The session list. Every session has its **own colour** (the dot on the left), kept on the map, in the table and in the charts. On the row: the pupil's name (and group, if any) and `<name> · <day> · n points · m m` (the day, the number of reads, the metres walked); `unknown day` if the session has no known day. **Swipe** a row sideways to remove it; `Session removed` appears with **Undo** for a few seconds. **⋮ → Clear all** deletes every imported session from the phone, after the confirmation `Delete all sessions imported on this phone?`.

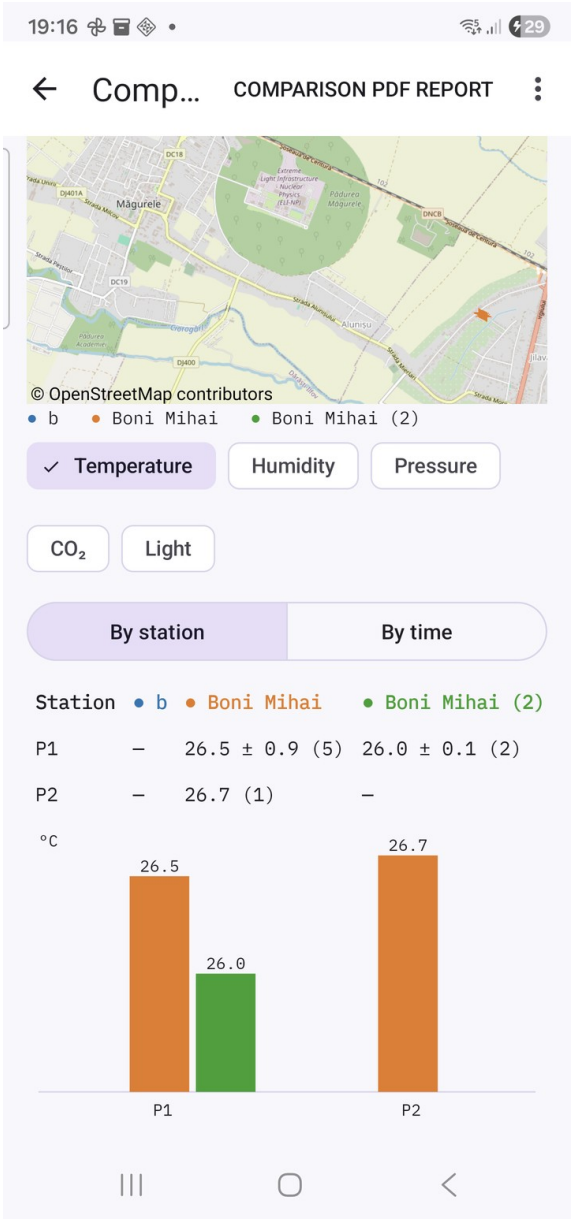
The map. The routes, points (small circles in the pupil's colour) and photographs (black squares) of every session. Tap a point: `<name> · <station> · <time>`; tap a square: the photograph, large. Satellite view follows the setting on the Map tab.

The sensor. The chips **Temperature**, **Humidity**, **Pressure**, **CO_2**, **Light** pick the sensor compared. The choice is remembered.

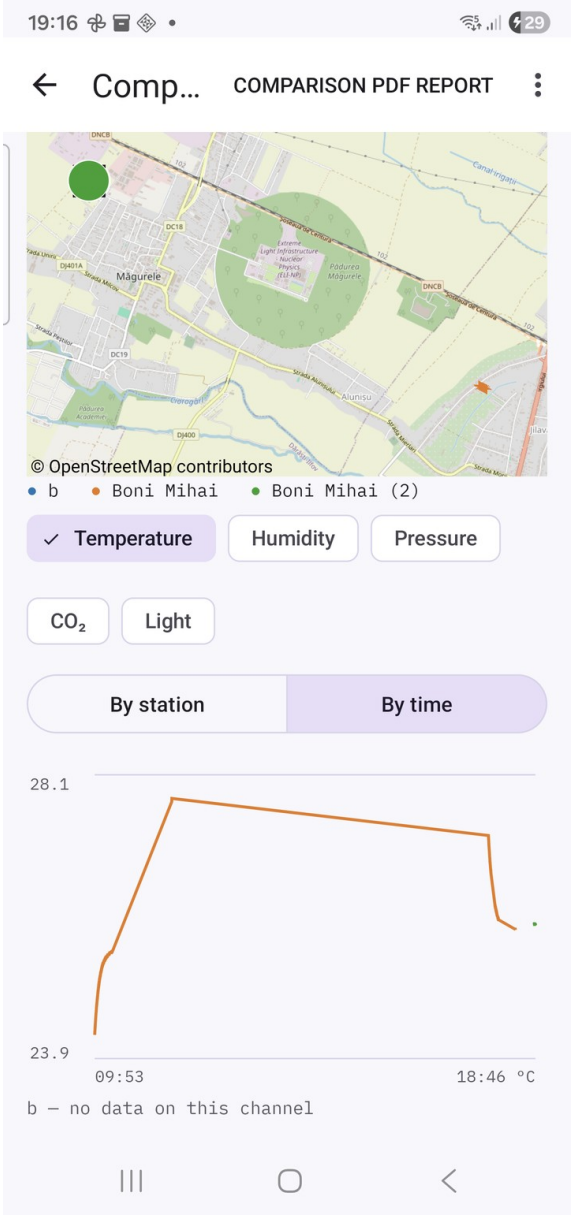
The two modes (the toggle above the chart):

- **By station** (default) — a table with one column per pupil and one row per station: the mean of the **Read** readings at that station, **± the standard deviation**, and **n** (how many reads). It works even if pupils walked on different days or different routes, as long as station names match. A bar chart shows the same data.

- **By time** — every pupil's 5-second series overlaid on a time-of-day axis. The axis is anchored to **the first loaded session's day**; anyone from another day is left out, with the reason under the chart: `Ana — different day`, `Ana — unknown day`, `Ana — no data on this channel`. If the sessions are not all from the same day, `"By time" needs sessions from the same day.` appears too.



The By station mode



The By time mode

⋮ → **Comparison PDF report.** The comparison report (chapter 7). It saves and opens the share sheet: `Report saved to Documents/FieldBox — opening the share menu.` Without sessions: `No sessions to compare.`; on failure: `Could not save the comparative report.`

Imported sessions stay in `Documents/FieldBox/compare/`, so the comparison survives closing the app — but **not** an uninstall: after reinstalling, the ZIPs must be sent again.

4.9 The notification

While a recording is running (after **Start**), the notification bar holds a permanent notification, visible on the lock screen too:

- **the title** `Field Box · 00:12:34` — the time elapsed since Start (measured by the box, so correct even after a pause of the phone);

- **the text** `P3 next · CO₂ 812 ppm · 4 points` — the current station, the CO_2 right now and how many reads were made today; or `box disconnected` if the box is not answering;
- the **Read now** button — a read at **the current station in the list**, exactly like the Read button on screen: its count grows, the phone vibrates. No need to open the app. Four taps at the same station are four reads at that station. The button is absent while the box is not answering;
- the **Stop route** button — stops **only the phone's GPS route** (the series and the notification); the recording stays on and the Done card does not appear. To end the walk use **Stop** on the Point tab.

The text refreshes every 5 seconds. The notification needs the **Notifications** permission (3.3).

4.10 The camera

Opens from the camera icon on the current row of the station list. The screen shows the camera preview and a single button, **Take photo**. There is no confirmation screen: if the shot is bad, take another. The back button closes the screen without a photo (except in the second the photo is being saved).

What happens on tap:

- the photograph is saved in the gallery, in the **DCIM/FieldBox** album, named `<BOX>_<station>_<day>_<time>.jpg` (for example `BOX_3C3C_P3_20260916_103015.jpg`; without a box, `NOBOX_…`);
- the GPS position and the time are written **into the photo itself**, so it places itself on a map in Google Photos or on a laptop;
- a row with the box, station, position, accuracy, date, time and file name is added to the day's photo list (`photos_<day>.csv`);
- the app confirms with `Saved BOX_3C3C_P3_20260916_103015.jpg`.

Messages: `This device has no camera.`, `Camera permission is needed to photograph a station.`, `The photo could not be taken. Try again.`, `The photograph was saved, but its manifest row was not:` … (the photo is in the gallery, but will not appear on the map; note it down).

5. A field day, step by step

The day before:

1. **Charge the box's powerbank** and the phone.
2. **Pair the phone with the box** over Bluetooth, if not done yet (3.4).
3. **Prepare the map, on Wi-Fi:** open the **Map** tab at home and zoom/pan over the whole area you will walk (in the street or satellite view, whichever you will use). The parts seen stay saved; in the field, with no signal, they are still drawn. Areas never seen show grey — the route and the points are drawn regardless.
4. **Fill in the profile** (name and class) — it goes into the report and the file names.
5. **Set the station plan** if you use names other than P1 … P5: from **edit** (Station plan) or from a CSV with **upload** (4.6).
6. **Grant the permissions** to the app, if it has not got them yet (3.3).

In the field:

1. **Power the box** (USB-C plug into the powerbank). The LED stays on for 2 seconds: the sensors are ready. Put the box in the backpack, in an airy spot — not in a closed pocket, so the CO_2 sensor can breathe.
2. Open the app, the **Point** tab, and tap your **`BOX_xxxx`** in the Box card. The strip at the top turns green. (The first time, the Profile card: name and class, or **Skip**.)
3. Check on **Boxes** that the clock does **not** say `(NOT SYNCED)`.
4. Get out under an open sky and wait for the pill to show `GPS n m`. From a cold start it can take 2–10 minutes; next to buildings, longer. It is not a fault. You can start without a fix too — it fills in on the way.
5. **Start**. The notification appears. **Box in the backpack, phone in the pocket** — within about 10 m of each other.
6. At every station: **tap its row** (it becomes current) → **Read** as many times as you like (wait for the buzz) → the camera if the station calls for a photo. Or, without opening the app: **Read now** from the notification. An unplanned station: **Other station…**
7. Glance at the strip at the top now and then: `link poor` or `Reconnecting…` = bring the box closer to the phone.
8. On return: **Stop**. The **Done** card appears.
9. Open **Map** once (so the map goes into the report), then **Files → PDF report** and **Send all**. Send the ZIP to the teacher (WhatsApp) or copy the folders to a laptop (chapter 6).
10. The next pupil with the same phone: **new** (or **New walk (next pupil)** on Done) → their name → **Start over** → from step 5.

The teacher, at home: collects the class's ZIPs in **Files → Compare** (or opens them straight from WhatsApp with Field Box), picks the sensor and the mode, then ⋮ **→ Comparison PDF report**.

6. The files

6.1 Where they are

Everything the app saves on the phone goes into a folder **named after the day**, so a day is easy to find and to copy or delete in one go:

What	Where
a day's files (points, track, series, photo list, session bundle, ZIP, report, logs downloaded from the box, that day's comparison PDF)	**Documents/FieldBox/<yyyy-MM-dd>/** — e.g. `Documents/FieldBox/2026-09-16/`
the photographs	**DCIM/FieldBox/** — visible in the Gallery
days archived with Start clean	**Documents/FieldBox/archive/ <yyyyMMdd>_<HHmm>/**
sessions imported into Compare	**Documents/FieldBox/compare/**
the plan template and a comparison PDF of mixed days	**Documents/FieldBox/** (the root)

At the first launch of version 0.8.0, files left by earlier versions directly in Documents/FieldBox are moved, once, into their day folders (by the date in each file's name). Files with no date in their name stay at the root. If a file could not be moved, the app tries again at the next launch.

6.2 Every file type

Every CSV file has a first line with the column names, a comma as separator and a dot as decimal separator — they open directly in Excel or LibreOffice.

The points (the phone's copy) — `points_phone_<yyyyMMdd>.csv`, one row for every Read that day. Columns: `date,time,station,lat,lon,temp_C,rh_pct,press_hPa,alt_m,co2_ppm,temp_scd_C,rh_scd_pct,light_lux` — date, time, station, latitude/longitude (empty without GPS), temperature, humidity, pressure, estimated altitude, CO_2, the temperature and humidity measured by the CO_2 sensor, light. The map and the report are made from this file. (After a reinstall of the app, variants `points_phone_<day> (1).csv` may exist too; all are read.)

The track — `track_<yyyyMMdd>_<HHmmss>.csv` (the time in the name = the time of Start), one row for every accepted GPS fix (every 5 seconds, if it moved at least 3 m and the accuracy is under 100 m). Columns: `date,time,lat,lon,accuracy_m`. A second Start on the same day makes a second file.

The phone's series — `log_phone_<yyyyMMdd>_<HHmmss>.csv`, the box's values read by the phone every 5 seconds between Start and Stop; the same columns as the points (station = the one set on the box). Gaps left by a link drop fill themselves from the box's memory when the link comes back.

The photo list — `photos_<yyyyMMdd>.csv`, one row per photograph. Columns: `box,station,lat,lon,acc_m,date,time,filename`.

The photographs — `<BOX>_<station>_<yyyyMMdd>_<HHmmss>.jpg` in DCIM/FieldBox, with the position and time written into the file.

The box's log (downloaded) — `BOX_xxxx_log_<yyyyMMdd>_<HHmmss>.csv` (on the box it is called `log_<day>_<time>.csv`; on download it gets the box's name in front). One row every 5 seconds from power-up; the same 13 columns as the points. The `station` column holds >REC at Start, <REC at Stop and the station's name on the row after a Read; otherwise it is empty.

The box's points (downloaded) — `BOX_xxxx_points_<yyyyMMdd>.csv`, one row for every Read the box accepted, the same columns. This is the official record; the phone's copy is for the map.

The session bundle — written at Stop: `session_<yyyyMMdd>_<BOX>_<Name>.json` (a summary: profile, box, plan, span, track, series, points, photos, distance) and `session_<yyyyMMdd>_<BOX>_<Name>.gpx` (the route, importable into Strava, Google Earth, QGIS). <Name> is the last word of the name, without diacritics; without a profile it is absent.

The report — `report_<yyyyMMdd>_<Name>_<Class>.pdf` (without a profile: `report_<yyyyMMdd>_<BOX>.pdf`). Contents: chapter 7.

The ZIP (Send all) — `session_<yyyyMMdd>_<BOX>_<Name>.zip`, holding: `session.json`, `route.gpx`, `report.pdf` (the report, under this fixed name), `photos/` (the day's photographs) and `box_log/` (the box's raw logs for that day).

The comparison report — `comparison_<yyyyMMdd>_<HHmm>.pdf` in the day folder (when every session is from the same day) or `comparison_mixt_<HHmm>.pdf` at the root.

The plan template — `plan_model.csv` at the root: P1 … P5, one station per line, no header.

6.3 How to copy a day to a laptop

Connect the phone to the laptop with the USB cable (choose "File transfer" on the phone) or use Google Drive / My Files. Copy the folder `Documents/FieldBox/<day>/` and, for the photos, `DCIM/FieldBox/`. Or, simpler, send the ZIP from **Send all** — it holds everything, photos and box log included.

6.4 How to delete a day

On the phone, in My Files: delete the folder `Documents/FieldBox/<day>/` (and the photos in DCIM/FieldBox, if you wish). The app has no button for deleting files from the phone; it only has **Start clean** (moves them to archive/) and **Delete** on Files (deletes from the **box**, not from the phone). Delete from the box only after the day's log has been fetched (PDF report or Send all with the box connected).

7. The reports

7.1 PDF report (the report of a day)

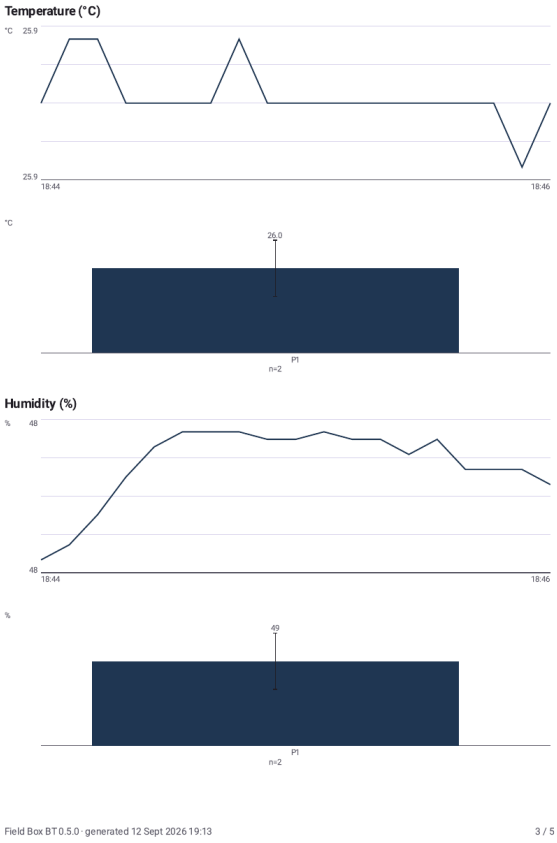
Made from **Files → PDF report** (today), **Files → Another day…** (another day) or from the **Done** card (without the map). A4 format. Sections without data are simply absent.


Field Box — BOX_6484 · Boni Mihai · 12 September 2026

18:44–18:45 · 0.03 km · 2 points at 1 stations · 19 series readings · 0 photographs

Station	Time	Lat	Lon	T °C	RH %	CO_2 ppm	hPa	Alt m	Lux
P1	19:04:27	44.35899	26.02280	25.9	49.2	1165	1007.5	48	1282
P1	19:13:15	44.35897	26.02291	26.1	49.0	1042	1007.6	48	1309

Field Box BT 0.5.0 · generated 12 Sept 2026 19:13 1 / 5



Field Box BT 0.5.0 · generated 12 Sept 2026 19:13 3 / 5

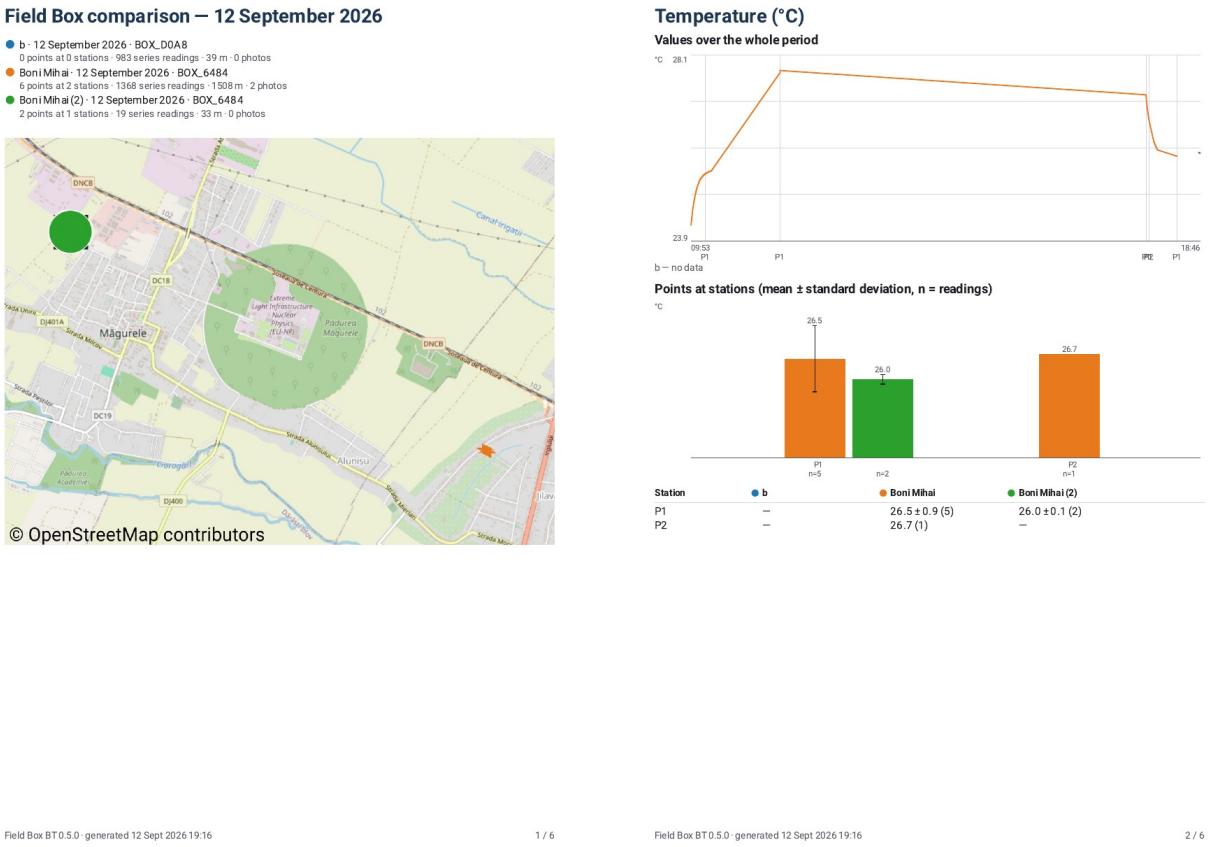
A page of the single-pupil PDF report

1. **The cover.** The title `Field Box — BOX_xxxx · <Name> · <date>`, a summary line built only from what exists (`10:29–12:41 · 1.84 km · 12 points at 5 stations · 1590 series readings · 3 photographs · box log 08:02–15:47 · 5570 readings`), the EASE mark, **the map** as last seen on the Map tab and, below, the first rows of the points table.
2. **The points table.** One row per Read: `Station`, `Time`, `Lat`, `Lon`, `T °C`, `RH %`, `CO₂ ppm`, hPa, `Alt m`, Lux. It continues on following pages (28 rows per page).
3. **`Summary by station (mean ± standard deviation, n = readings)`.** One row per station, one column per sensor: the mean of the reads at that station, ± the standard deviation, and the number of reads. Present when there are points.
4. **The chart pages** — `Values over the whole period`: for every sensor, the trend over the whole period (from the phone's series) with the station moments marked, plus the per-station bars with error bars; two sensors per page. These pages appear **only** when the box's log is not on the phone (otherwise the Box day charts replace them).
5. **Box day** (when the box's log is on the phone; one set per box): the page `Box day — BOX_xxxx · <date> · 08:02–15:47`, with **the day's colour bar** (every 5 seconds coloured by its CO_2 band), the curve `CO₂ (ppm) over the day` on the lesson's bands, the band legend (plus `Recording` = the grey span between Start and Stop), **the day's facts** (`Highest CO₂ … ppm at hh:mm · lowest … ppm at hh:mm`, `Air was excellent or good n % of the logged time · …` with the episodes over 1200 ppm, `Warmest … °C at hh:mm · coolest …`, `Brightest … lx at hh:mm`, `Pressure …–…` hPa) and a table `Channel · min (at) · max (at) · mean · median · sd · n`; the **24 h chart pages**, two sensors per page, with the recording span shaded (`shaded = recording hh:mm–hh:mm`); the **`Data quality — BOX_xxxx`** page: `Coverage hh:mm–hh:mm · n rows · n % of the day`, `Gaps: …` (the gaps, with their length), `Restarts: …` (the box's restarts), `Files: …` (the source files), `Firmware … · Field Box BT 0.11.0`. On a day with no points and no map, the Box day page becomes the cover itself.
6. **The photographs**, 6 per page, each with the station, the time and the matched reading printed underneath (the nearest point or, failing that, the series row within 5 minutes).

The footer of every page: `Field Box BT 0.11.0 · generated <date time>` and the page number.

7.2 Comparison PDF report

From **Compare → ⋮ → Comparison PDF report**.



A page of the comparison PDF report

1. **The cover** `Field Box comparison — <day>` (or `different days`) with the legend (colour, pupil, day, box; points, readings, metres, photos per pupil) and the map with every route.
2. **One page per sensor** (Temperature, Humidity, Pressure, CO_2, Light): `Values over the whole period` — every pupil overlaid on the time axis (anchored to the first session's day; the others are left out, with the reason) — and `Points at stations (mean ± standard deviation, n = readings)`: the table and the per-station bars.

7.3 The ZIP (Send all)

One file to hand to the teacher: `session.json`, `route.gpx`, `report.pdf`, `photos/…`, `box_log/…` (6.2). Made from the Done card or from Files; with the box connected it first fetches the day's log. The Compare screen reads exactly this ZIP.

8. Troubleshooting

8.1 The box and the link

What you see	Probable cause	What to do
the box is missing from `Choose a box` / the Box card	it was never paired with this phone	**Settings → Bluetooth → Pair new device → `BOX_xxxx`** (the box must be powered), then back to the app
`No paired box found. …`	same, no box paired at all	the **Bluetooth settings** button in the message
`BOX_xxxx` does not appear even in Bluetooth settings	the box is not powered, or already connected to another phone	check the red light; disconnect the other phone (**Disconnect**)
`The box is not answering. Check it is switched on and within a few metres, then tap Connect again.`	the box is off, too far, or busy with another phone	come within a few metres, check the power, tap **Connect to a box** again; if it persists, unplug and replug the box's cable
`Bluetooth is switched off on this device. …`	the phone's Bluetooth is off	turn it on in the quick settings
`Could not connect. Try again; if it keeps failing, switch the box off and on.`	a passing error	once more; then the box unplugged and plugged back in
`Bluetooth permission is required to talk to the box. …`	the Nearby devices permission is missing	**Settings → Apps → Field Box BT → Permissions**
`Reconnecting to BOX_xxxx… / Link to the box dropped — reconnecting…`	the link dropped (range, an obstacle)	nothing — the app restores it by itself; bring the box closer; the box keeps logging meanwhile
🟠 `link poor` often	at the edge of range (~10 m), the box in a metallised pocket, the phone in another bag	the box in the backpack of whoever holds the phone
🟣 `link none` for more than a few seconds	the link was just made and is not answering	wait; if it stays, **Disconnect** and reconnect
the link drops every few minutes	the phone is saving power	turn off battery optimisation for Field Box BT in Android settings; leave the link notification in place
the clock says `(NOT SYNCED)`	the box has just started and has not received the time	**Disconnect**, then connect again; if it persists, the box's cable out and in
`Recording: no` although you pressed Start	the box was restarted (cable) after Start	press **Stop**, then **Start**
the box's LED does not light at all	no power, loose cable, flat powerbank	push the plug in; another cable / another powerbank
the LED blinks very fast	storage full	see 8.6

8.2 GPS and the route

What you see	Probable cause	What to do
the pill says `no GPS` for minutes, outdoors	a cold start of the receiver	open sky, away from buildings, screen on on Point; 2–10 minutes; not a fault; you can work anyway (reads are saved without coordinates)
`no GPS` indoors	there is no GPS signal inside buildings	normal; reads work, without coordinates; the map shows `n points with no GPS do not appear on the map`
the amber pill `GPS n m (weak)`	accuracy over 15 m	usable; it improves while walking under an open sky
`Location is switched off on this device. …`	the phone's Location is off	turn it on in the quick settings
`Location permission refused. …`	you refused the permission	**Settings → Apps → Field Box BT → Permissions → Location → Allow (precise)**
`Route: not running — check the Location permission…`	the permission was missing at Start	tap the message, grant the permission, then **Stop** and **Start**
`Route: not running on the phone (the box is recording) — tap here…`	the app was updated or closed by Android during the walk	**tap the message**: the route resumes in a new file; nothing lost on the box
`Route: no point yet · n GPS fixes rejected (last ±x m, limit 100 m)`	the positions received are too inaccurate (indoors, a closed courtyard)	go to open ground; it resolves by itself
`Route: n points · m skipped (weak GPS)` keeps growing	indoors, under dense trees, standing still	normal; "skipped" is not lost data but rejected positions
`Route: off — this device has no GPS receiver`	a tablet without GPS	there will be no route; Read, Stop, report and ZIP still work
the route has "jumps" on the map	inaccurate positions accepted (up to 100 m)	normal next to tall buildings; the curve passes through every accepted position

8.3 Reads, stations, photographs

What you see	Probable cause	What to do
Start is greyed out	no box connected, or you are working without a box	connect the box (the Box card or the Boxes tab)
Read is greyed out	same	same; photographs work anyway
`The clock is not set yet.` at Start	the box has no time	**Disconnect** and reconnect (the time is sent on connection)
`The storage is full.` at Start	the box's storage is full	see 8.6
`The box refused the request.`	the box answered with an error	reconnect; if it persists, the box's cable out and in
`Could not reach the box.` at Start or Read	the link dropped just then	wait for the reconnection (green strip at the top), tap again
`The box could not record this point — check the clock is synced and there is free storage.`	the box refused the Read: clock not set or storage full	check the clock and `Free storage` on Boxes

What you see	Probable cause	What to do
Stop does not appear	the recording is not running (Start was not pressed or was refused)	press **Start**; watch the status line for a refusal
Stop does not appear on a tablet without GPS	—	in 0.8.0 Stop works without GPS too; if it is missing, see the case above
a station's count does not grow after Read now from the notification	the box was not answering (`box disconnected`) at that moment	the Read now button is absent while the box is not answering; try again when the notification shows CO_2 again
Read now from the notification counted at another station	the current station in the list was another one	tap the row of the station you are at before Read now
the notification is missing	the Notifications permission was refused, or the recording is not running	grant the permission; press Start
`Camera permission is needed to photograph a station.`	the Camera permission is missing	from Settings → Apps → Field Box BT
`The photo could not be taken. Try again.`	the camera is busy with another app or failed	close other camera apps; try again
`The photograph was saved, but its manifest row was not: …`	the photo is in the gallery, but could not be added to the day's list	note the station and time; the photo will not appear on the map/report
on the map, a photo shows `the photo was not found`	the photo was deleted or moved out of DCIM/FieldBox	look for it in the gallery; put it back into DCIM/FieldBox
a station added by mistake with Other station…	—	long press on the row → **Remove** (its reads stay in the files)
values show – on Live / in a bubble	the sensor has not produced a value yet (CO_2 needs a few seconds after power-up) or a sensor is not answering	wait 30 s; if one sensor stays – all day, note which box it is: that sensor needs checking
the arrows are missing (only …)	the first 40 s after connecting, or the link had a gap	wait; they come back by themselves
the Live charts stay at `waiting for data`	the box has firmware older than 0.3.1-bt, or you have just connected	a few seconds; if it persists, update the box (installation guide, chapter 5) — values and reads work anyway
a `Field Box` dialog with a cheerful message	you passed 1/5/10/20/40 reads today	tap OK; it affects nothing

8.4 The app, updates, crashes

What you see	Probable cause	What to do
the app closed suddenly (crash) during the walk	—	reopen it; **the data is on disk**: every Read, every position and every photo is written to the files immediately. If Live shows `Route: not running on the phone…`, tap the message

What you see	Probable cause	What to do
after an app update the counts are at zero	the on-screen counts belong to the current day; an update can reset them	the files are intact; the map and the report show every read; tap the Route message on Live so the route resumes
on a new day the station list is at zero	normal: counts are per day	nothing
the phone drains fast	GPS every 5 s + permanent Bluetooth + the screen on	screen off in the pocket (Read now from the notification); lower brightness; a powerbank for the phone on long walks
the notification `Field Box — connected to BOX_xxxx` does not go away	normal while the box is connected	**Disconnect** on Boxes if you want it gone
the app asks for permissions again at launch	one of them is missing	grant it; Android stops asking after several refusals — then from Settings

8.5 Files and reports

What you see	Probable cause	What to do
the files are not where I expected (directly in Documents/FieldBox)	since 0.8.0 every day has its own folder	look in `Documents/FieldBox/<yyyy-MM-dd>/`; old ones were moved there automatically
today's map / report are empty although I measured	you pressed **Start clean** — the files are in `archive/`	copy them back into the day folder in My Files, or make the report on a laptop from the archive
`Nothing to report yet.` / `Nothing to report for <day>.`	no points, no series, no photos and no box log for that day	connect the box (the report fetches the log) or pick another day
Another day… shows `Nothing on the phone to report yet.`	no day with points, series or a box log	connect the box and make a **PDF report** — it fetches the day's log
Another day… does not list a day I only have a track for	tracks and photos alone do not make a day "with data"	was Send all done that day? The ZIP has everything
`The report could not be created`	not enough space on the phone, or a broken file	free up space; try again; if it persists, copy the day folder to a laptop
`The package could not be created` (Send all)	same	same
the report has no map	the report was made from the Done card, or Map was not opened today	open **Map**, then **Files → PDF report**
the report has no Box day pages	the box's log is not on the phone	make the report with the box connected; phone next to the box while it downloads
`Downloading the box log…` takes long	Bluetooth is slow: a whole day is a few hundred kB	~20 s; keep the phone next to the box
`This file is being recorded — stop the recording first.`	you tried to delete the open log	**Stop** on Point, then Delete

What you see	Probable cause	What to do
`Could not delete the file.`	the box did not answer	Refresh, once more
`Deleted — the box opened a fresh log (it logs whenever it is powered).`	you deleted the current log; the box opened another	normal
Send all does not open WhatsApp	the share sheet was dismissed	`Package saved` → **Send**; or find the ZIP in the day folder
`Plan "x" loaded: 1 stations` **although the file had many**	the file was saved with another separator (tab)	save from Excel as CSV (comma), or one station per line
`No stations found in x.csv.`	an empty file or header only	fill in one station per line
`Could not read the file.`	the file could not be opened from the source app	copy it into My Files first

8.6 The box's storage

What you see	Probable cause	What to do
`n kB free on the box — running low; download and delete old files on Files`	under 300 kB free	make a **PDF report** or **Send all** (fetches today's log), then **Files → Delete** on the old logs
`n kB free on the box — full; the box has stopped logging` / `The storage is full.` / **very fast LED**	no space and no old logs to delete	download and delete on Files; or delete an old points file (they are never auto-deleted); the box resumes its log by itself once there is room
old logs vanished from the box	the box deletes its oldest log by itself below 100 kB free	normal; download logs in time (the day's report)
the box's files vanished after a firmware update	the update from 0.2.0-bt to 0.3.x reformats the storage	expected; download everything before updating

8.7 Compare

What you see	Probable cause	What to do
`Maximum 6 sessions.`	a 7th session	swipe one out of the list (or **Clear all**)
`x is not a Field Box session.`	the file is not a ZIP from Send all or a session_….json	ask for the right ZIP
`Could not import x.`	a broken or incomplete ZIP (WhatsApp truncated it)	have it sent again
`Ana — different day` **under the By time chart**	the session is from another day than the first loaded	normal for By time; use **By station** for cross-day comparisons; or load the session from the wanted day first
`Ana — unknown day`	an old bundle, with no day	By station works
`Ana — no data on this channel`	that session lacks the chosen sensor	pick another sensor
`"By time" needs sessions from the same day.`	mixed days	same
the sessions vanished after a reinstall	the compare/ folder belongs to the app	send the ZIPs again

8.8 How to start over

What you want	What to do	What is lost
another pupil, same day, same files	**new** → **Start over**	only the on-screen counts
a new session on the same day, with an empty map	**Map** → ⋮ → **Start clean**	nothing: the files go to `archive/`
deleting a day from the phone	My Files: the folder `Documents/FieldBox/<day>/`	that day (irreversible)
an empty box for tomorrow	**Files** → **Delete** on every file (after downloading)	the files on the box (irreversible)
the app as new	uninstall + reinstall	the profile, the plans, the Compare sessions; the files in Documents/FieldBox and the photos stay

If the link keeps dropping, note what **Link drops** says (long press on the versions line, Boxes tab) before restarting anything — that list is what to send when you ask for help, together with the `App v…` · `firmware` v… line.

9. Specifications

Sensors	BME280 (temperature, humidity, pressure, estimated altitude), SCD41 (CO_2), VEML7700 (light)
The box's log	one row every 5 s, from power-up; a new file at every power-up and at midnight
The box's points	one row per Read; one file per day; never auto-deleted
The box's storage	~2.5 MB (~150 h of log); below 100 kB free it deletes the oldest log; the app warns below 300 kB
History for Live	the last 50 minutes (600 samples at 5 s), sent on connection
The phone's series	one row every 5 s between Start and Stop; gaps filled from the box's history
GPS	one fix per second for the pill; the track at 5 s, accepted if accuracy ≤ 100 m and movement ≥ 3 m; pill green ≤ 15 m, amber above
Read	instant: the values at that moment + the latest GPS fix
Trend arrows	mean of the last 10 s against the mean 30–40 s ago; dead bands 30 ppm / 0.2 °C / 1.5 % / 0.5 hPa / 10 lx or 20 %; unknown for the first ~40 s
Link quality	the last 10 answers: `good` median < 0.3 s; `poor` one answer > 1 s or missing; `ok` otherwise
CO_2 bands	< 800 Excellent · 800–1200 Good · 1200–2000 Moderate · 2000–5000 Poor · ≥ 5000 Dangerous
Temperature bands	< 5 Cold · 5–15 Cool · 15–28 Comfortable · 28–35 Hot · ≥ 35 Very hot (°C)

Bluetooth	Bluetooth Classic (serial port); name B0X_xxxx; practical range ~10 m; paired once
The box's clock	no battery; set by the phone on connection; at most one hour of rows kept until set
LED	solid 2 s at boot = sensors OK; 0.5 s blink = recording; 0.1 s blink = storage full
Android	10 or newer; Location (precise), Bluetooth/Nearby devices, Camera, Notifications
Files	CSV (comma, decimal point, header), JSON, GPX, PDF (A4), ZIP, JPG with position in EXIF
Folders	Documents/FieldBox/<yyyy-MM-dd>/, archive/, compare/; DCIM/FieldBox
Compare	at most 6 sessions; By time needs the same day
Plans	at most 200 stations from a CSV; duplicate names removed
Versions	app 0.11.0; firmware 0.4.1-bt
Compatibility	0.2.0-bt boxes work: the log runs only between Start and Stop, and the Live charts may stay empty; updating to 0.3.x reformats the box's storage (download everything first)

10. Glossary

- **Box** — the logger with sensors, `B0X_xxxx`.
- **Station** — a pre-arranged place where the group stops and reads; it has a name (P1, P2 … or any text).
- **Plan (station plan)** — the ordered station list of a walk; it has a name; `Standard plan` = P1 … P5.
- **Walk** — one pupil's pass along the route, from Start to Stop; **new** begins the next walk.
- **Read** — one press of **Read** (or **Read now**): all the sensor values at that moment, plus the GPS position, at the current station.
- **The current station** — the highlighted row of the list; Read and Read now go there.
- **Point** — a read, as it appears in the files, on the map and in the report.
- **Series** — the values taken automatically every 5 seconds (the phone's: between Start and Stop; the box's: all the time).
- **Log** — the box's continuous file, one row every 5 seconds.
- **Track / route** — the GPS positions recorded by the phone every 5 seconds.
- **GPS fix** — one determination of the position by the GPS receiver, with an accuracy in metres.
- **Session** — everything that came out of one day (track, points, series, photos), summarised in `session_….json`.
- **Bundle / ZIP** — the file from **Send all**, with everything belonging to the session.
- **Day folder** — `Documents/FieldBox/<yyyy-MM-dd>/`.
- **Archive** — the folder **Start clean** moves a day's files into.
- **Day report** — the PDF with points, map, charts and the Box day pages.
- **Box day** — the report pages made from the box's continuous log.
- **Link** — the phone–box Bluetooth connection; `good` / `ok` / `poor` / `none`.
- **Drop** — an interruption of the link, restored automatically.

- **NOT SYNCED** — the box's clock has not yet been set by the phone.
- **Pairing** — the one-time Bluetooth pairing, from the phone's settings.
- **Band** — a range of values with a name and a colour (Excellent … Dangerous; Cold … Very hot).
- **ppm** — parts per million (CO_2); **hPa** — hectopascal (pressure); **lx** — lux (light).
- **Manifest (photos list)** — the file `photos_<day>.csv` that links every photo to its station and position.

11. Field card (one-page summary)

1. Box powered → red light, LED solid 2 s → into the backpack.
2. **Point** tab → tap your `BOX_xxxx` (first time: the profile or **Skip**) → green strip at the top. On **Boxes**: the clock without `(NOT SYNCED)`.
3. **Box in the backpack, phone in the pocket** — within ~10 m.
4. Outside, under an open sky, until the pill reads `GPS n m` (2–10 min from cold).
5. **Start**. The notification `Field Box · 00:00:00` appears.
6. At every station: **tap its row** → **Read** (as often as you like; wait for the buzz) → the camera if needed. Or **Read now** from the notification, without opening the app. An unplanned station: **Other station…**
7. `link poor` / `Reconnecting…` → bring the box closer to the phone.
8. On return: **Stop** → the **Done** card.
9. **Map** (once) → **Files → PDF report** → **Send all** → the ZIP to the teacher (WhatsApp), or copy `Documents/FieldBox/<day>/` and `DCIM/FieldBox` to a laptop.
10. The next pupil: **new** → their name → **Start over** → from step 5.
11. Box storage running low (`running low`): report made, then **Files → Delete** on the old logs.
12. The teacher: **Files → Compare** → **Add sessions** (the class's ZIPs) → ⋮ **→ Comparison PDF report**.